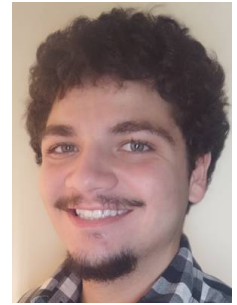


JÚLIO O. AMANDO DE BARROS

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- 🎓 Google Scholar Profile
- in LinkedIn
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- 🌐 Personal Website



About

Physicist and materials scientist (ongoing Ph.D., ETH Zürich; M.Sc., USP Brazil) with expertise in data-driven modelling, scientific computing, and experimental design. Skilled in Python (data analysis, visualization, statistical modelling), Abaqus (mechanical modelling), GROMACS (molecular modelling), and LabVIEW (automation), with hands-on experience in microscale material preparation and mechanical testing. Experienced in bridging theory and experiment by developing custom data acquisition pipelines, automating analysis, and designing novel experimental setups. Passionate about applying modelling and advanced data analytics to support innovation in R&D and connect material science to real-world applications.

Work Experience

- MAR 2022 – CURRENT **ETH Zürich** Ph.D. Candidate/Research Assistant
- Developed mechanistic models linking wood anatomy and moisture gradients to predict rheological behavior.
 - Applied micromechanical techniques to test the wood cell wall behavior, we identified crucial error sources of the method, and established protocols to correctly characterize structured organic materials.
 - Designed and built a custom microtensile testing system with automated climate and load control, improving reproducibility of material testing.
 - Supervised bachelor's thesis and supported computational teaching in Python and FEM.
- JUL 2019 – DEZ 2021 **University of São Paulo** M.Sc. in Physics
- Applied molecular dynamics and ab initio simulations to cellulose-ion and cellulose-drug interactions, supporting drug delivery and energy harvesting applications.
 - Combined ab-initio and Molecular Dynamics simulations to explore functionalized cellulose fibrils and amino acids interactions for controlled release purposes.
 - Teaching assistance on Experimental Physics and Physics of Heat.
- FEV 2016 – JUL 2019 **University of São Paulo** Undergraduate Research Program
- Performed Dynamical Mechanical Analysis on Brazilian hardwoods, analysing their frequency and temperature dependence.
 - Performed quantitative wood anatomy characterization and compared mechanical and anatomical properties using dimensionality reduction and clustering techniques to identify key features.

Education

MAR 2022 – CURRENT	ETH Zurich Ph.D. in WOOD-WATER INTERACTIONS FROM A MICROMECHANICAL PERSPECTIVE Advisor: Prof. Ingo Burgert and Dr. Falk K. Wittel
JUL 2019 – DEC 2021	University of São Paulo M.Sc. in Physics - THEORETICAL STUDIES ON CELLULOSE FIBER INTERACTIONS WITH DRUGS AND IONS Advisor: Prof. Kaline Coutinho
FEV 2014 – FEV 2019	University of São Paulo B.Sc. in Physics

Technical Strengths

Programming Languages	Python	●●●●●○
	C++, Bash, MATLAB	●●○○○○
Software Skills	Linux, SSH, Git, ABAQUS, LabVIEW, Arduino, GROMACS, Gaussian, Mathematica, Blender, Illustrator, AEM Website Manager, MS Office. Python Libraries: Numpy, Scipy, Pandas, Scikit learn, Matplotlib, Plotly, OpenCV, PyTorch.	
Experimental Experience	Optical and Electron Microscopy, Focused Electron and Ion Beam techniques (FIB and FEBID), Nanoindentation, Microtensile Testing, Dynamic Mechanical Analysis (DMA), Microtoming, Ultramicrotoming, and other biological sample preparation techniques.	

Other Information

Languages	Portuguese	Native
	English and Spanish	Advanced
	German	Intermediate (B1)
	French	Basic

Highlighted Teaching assistant roles

- Materials in construction: Finite Element Methods Lab. (ABAQUS and Python) (2024-2025) (Dr. Falk Wittel).
- Computational Scientific Investigation (Python) (2023-2025) (Prof. David Kammer and Dr. Falk Wittel).

Highlighted academic events

- Oral Presentations in Computational Methods in Wood Mechanics - ECCOMAS (2023 and 2025)
- English Courses for Presentations and Writing (2019-2024)
- *Machine Learning Techniques Applied to Computational Materials Science with Hands-on* (2020)

Other engagements

- Musician and member of the executive board of an association to organise and foster musical events in Zürich.
- Outdoor activities, such as white water kayaking and skiing.

Selected Publications

Amando de Barros, J., Wittel, F. Unifying model for the rheological behavior of hygroresponsive materials. *Physical Review E*. 2024; 109(4):044139. <https://doi.org/10.1103/PhysRevE.109.044139>

Amando de Barros, J., Schwiedrzik J., Wittel F. Resolving Discrepancies in Wood Micromechanics: Strain-Mapped Compression of Tracheid Wall Micropillars. *Composites Part A*. 2025. <https://doi.org/10.1016/j.compositesa.2025.109209>